SUBJECT-LEVEL STUDENT ACADEMIC PERFORMANCE SEGMENTATION USING STABILITY-AWARE K-MEANS++

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Abstract—Student achievement is commonly summarized through aggregate scores, although students with similar overall performance may exhibit different subject-level strengths and difficulties. This study develops a subject-level segmentation pipeline using K-Means++ on Penilaian Akhir Semester (PAS) records from SMA Negeri 1 Jakenan. The supplied workbook contains 19,401 records, 1,292 unique student names, 38 class labels, and 30 subject labels. After class-label normalization, record cleaning, curriculum-compatible cohort construction, subject-coverage filtering, student-completeness filtering, cohort-specific median imputation, zero-variance feature removal, and z-score standardization, 1,277 students were retained across five cohorts. K-Means++ was evaluated for k=2 - 8 using 30 random seeds per configuration. Silhouette coefficient, Davies - Bouldin index (DBI), and pairwise Adjusted Rand Index (ARI) were combined through Pareto-based multi-objective selection. The selected solution was k=2 for every cohort, with mean Silhouette values of 0.2299 - 0.3165, DBI values of 1.1787 - 1.5038, and mean pairwise ARI values of 0.9498 - 1.0000. A supplementary benchmark implemented random-initialized K-Means, DBSCAN, BIRCH, Gaussian Mixture Models (GMM), and Mean Shift on the same standardized cohort matrices. DBSCAN produced high Silhouette values in some settings but excluded 72.5% - 95.1% of observations as noise, limiting direct comparability. The results indicate repeatable initialization-level partitions with moderate separation, rather than definitive or universally valid student categories. The resulting profiles may support exploratory subject-level academic monitoring when interpreted alongside teacher judgment and contextual evidence.

Index Terms—educational data mining, academic performance, student segmentation, subject-level scores, K-Means++, clustering stability, Silhouette coefficient, Adjusted Rand Index.

I. INTRODUCTION

Educational institutions routinely accumulate assessment records that can be examined to understand learning outcomes and identify patterns that are not apparent in routine reporting. Educational data mining (EDM) applies computational and statistical techniques to educational data to derive information that may support learners, educators, and institutions. Dutt et al. [1] describe EDM as a growing research area and review the use of clustering and other data-mining methods in educational settings. Such methods are particularly relevant when educational data contain multiple dimensions and the objective is exploratory rather than the prediction of a predefined label.

Academic performance is multidimensional. A student may obtain a moderate overall score while performing relatively well in some subjects and experiencing difficulty in others. If analysis uses only an overall mean or median, the differences among subject domains are compressed into a single value. Chen et al. [2] similarly emphasize that the use of overall grades may overlook individual subject characteristics. Subject-level features can preserve more information for profiling, provided that the analysis respects differences in curriculum and subject availability.

Prior EDM studies have examined academic performance through multiple analytical perspectives. Rahman et al. [3] combine programming-practice indicators and academic scores using clustering and association analysis. Alhazmi and Sheneamer [4] examine clustering and classification for early performance analysis in higher education, while Qin et al. [5] develop an early-warning model integrating subjective and objective elements. Vives et al. [6] investigate sequence-based prediction using LSTM networks. These works demonstrate the breadth of data-driven academic analysis, but their predictive objectives differ from the unsupervised descriptive segmentation considered here.

Clustering is useful when the research objective is to discover groups without prespecified class labels. However, the resulting partitions depend on feature representation, preprocessing, distance geometry, algorithm configuration, and initialization. Pecuchova and Drlik [7] demonstrate the use of clustering on students' digital traces and highlight the value of temporal and content-specific analysis. Chen et al. [8] use versioned repositories to derive learning-behavior indicators. These studies motivate careful consideration of the data source: the present work uses examination scores rather than behavioral traces, and therefore describes academic outcome profiles rather than engagement or learning processes.

K-Means is a widely used partitioning method, but its solution can vary with centroid initialization. Harris and de Amorim [9] provide an empirical comparison of K-Means initialization algorithms across synthetic and real-world datasets, demonstrating that initialization choice is an important experimental dimension. In educational data mining, Andre et al. [10] compare K-Means, DBSCAN, BIRCH, Gaussian Mixture Models, and Mean Shift for clustering undergraduate thesis-topic data. Mutka et al. [11] further show how EDM can support student-retention

analysis through detailed learning-validation indicators. These studies provide the methodological context for repeated K-Means++ runs and the supplementary comparison of rival algorithm families in this paper.

This study analyzes PAS subject scores from SMA Negeri 1 Jakenan and addresses three research questions: (1) How can long-format PAS records be transformed into curriculum-compatible student-by-subject matrices? (2) Which number of clusters provides the best observed balance of internal cluster quality and initialization repeatability for each cohort? (3) What subject-level score profiles characterize the selected partitions? The contribution is a reproducible preprocessing and segmentation workflow, repeated K-Means++ evaluation across 30 seeds, Pareto-based selection using three internal criteria, and a transparent exploratory comparison with five rival algorithm families. The study does not claim to predict future performance or establish causal explanations for student achievement.

II. RELATED WORK

EDM provides a framework for analyzing data generated in educational environments. The systematic review by Dutt et al. [1] discusses the development of EDM and the application of clustering to educational datasets. Clustering can support the identification of groups with similar characteristics, but its usefulness depends on whether the selected attributes represent the educational question. In the current study, the attributes are individual PAS subject scores, not demographic characteristics, attendance, or online activity.

Several studies use academic outcomes as input to predictive or descriptive models. Rahman et al. [3] investigate practical programming skills alongside academic performance, illustrating the value of combining different evidence sources. Alhazmi and Sheneamer [4] analyze early academic indicators and compare clustering and classification techniques. Qin et al. [5] integrate subjective and objective factors in a performance early-warning model. Vives et al. [6] apply LSTM-based models to sequential assessment information in programming fundamentals. These approaches are relevant to the broader goal of academic monitoring, but they require different data structures and evaluation protocols from unsupervised clustering of a single semester's subject grades.

Student profiling has also been studied using non-grade educational traces. Pecuchova and Drlik [7] examine Moodle digital traces and compare clustering approaches for identifying student activity profiles associated with dropout risk. Chen et al. [8] mine version-control repositories to derive learning-behavior indicators from programming assignments. Mutka et al. [11] use detailed programming-process indicators for early outcome prediction. Such work suggests that academic monitoring can benefit from richer data, while also clarifying the scope of the present study: the available PAS workbook contains subject scores, so the analysis cannot infer attendance, effort, motivation, or temporal learning behavior.

Algorithm selection and validation are central to clustering. Harris and de Amorim [9] compare 17 initialization algorithms and show that initialization performance depends on data characteristics. Andre et al. [10] compare multiple clustering families in an EDM context, providing a practical precedent for evaluating partitioning, density-based, hierarchical/CF-tree, and probabilistic approaches on educational data. The present study adopts K-Means++ as its primary method because the research design explicitly evaluates its initialization repeatability, and uses the rival families discussed by Andre et al. as a supplementary benchmark. The benchmark is performed on the current PAS matrices and does not reproduce Andre et al.'s dataset or experimental settings.

The research gap addressed here is narrower than a general claim that clustering is underused in education. Existing studies establish the value of EDM, student profiling, and comparative clustering, but the supplied research set does not provide a direct analysis of subject-level PAS profiles within curriculum-compatible cohorts at SMA Negeri 1 Jakenan using repeated K-Means++ initialization and a joint internal-quality/repeatability selection rule. This study addresses that specific methodological and contextual gap. It does not claim that the resulting two groups are naturally occurring student types or that the method outperforms all alternatives in general.

III. DATASET AND DATA AUDIT

The dataset is the PAS Semester 1 workbook for SMA Negeri 1 Jakenan. Sheet1 contains 19,401 long-format records and eight fields: ABSEN, KELAS, NAMA, NILAI, MATA PELAJARAN, GURU PENGAMPU, NOMOR, and HARI KE. The initial audit identified 1,292 unique student names, 38 raw class labels, and 30 subject labels. The workbook also contains a second sheet with 21 rows of auxiliary entries; the executed Colab workflow reads Sheet1 as its analytical input.

The numeric conversion audit identified 243 nonnumeric NILAI entries: 221 entries marked “Belum Ikut” and 22 entries marked “Kristen.” The “Belum Ikut” marker represents an uncompleted or unavailable assessment and was treated as missing rather than converted to zero. The 22 “Kristen” markers occurred in repeated student - subject pairs for which a numeric counterpart was present. The preprocessing prioritizes the numeric value and retains one record per normalized student - subject key. This is a data-cleaning decision that should be confirmed with the school data owner before operational deployment.

Class labels “XII - IPA n” were normalized to “XII - MIPA n” before cohort assignment. The student key used for linkage is normalized class plus student name; it is not included as a clustering feature. The cleaned long-format table contains 19,379 records. Five curriculum-compatible cohorts were then constructed: Grade X, Grade XI classes 01 - 04, Grade XI classes 05 - 12, Grade XII MIPA, and Grade XII IPS. This grouping follows the observed subject-offering pattern in the workbook and avoids treating a subject not offered to a cohort as a zero score.

TABLE I. RAW DATA AUDIT

Audit item	Observed value
PAS records in Sheet1	19,401
Unique student names	1,292
Raw class labels / subject labels	38 / 30
Nonnumeric NILAI entries	243 (221 “Belum Ikut”; 22 “Kristen”)
Repeated student - subject pairs	22 pairs (44 rows)
Records after cleaning	19,379

IV. METHODOLOGY

A. Subject-Level Feature Construction

Within each cohort, the long-format data are pivoted into a matrix X whose rows represent students and whose columns represent subjects. A cell X_{ij} contains the PAS score of student i in subject j. The student key is used only as the row index. It is excluded from the numerical feature space so that identity or class code cannot influence distance calculations.

Subject coverage is calculated as the proportion of students in a cohort with a numeric score for that subject. The primary experiment retains a subject when its coverage is at least 80%. After selecting the subject features, a student is retained when at least 80% of those features are observed. The two thresholds address different sources of incompleteness: subject-level sparsity and student-level missingness. The retained matrices are constructed independently for each cohort.

B. Missing-Value Treatment and Standardization

Remaining missing cells are imputed using the median score of the corresponding subject within the same cohort. Cohort-specific imputation avoids borrowing score distributions across grade levels or academic streams. Features with zero variance after imputation are removed because they cannot distinguish observations in the cohort. The remaining features are transformed using z-score standardization, implemented with StandardScaler. Standardization gives each subject feature a comparable scale for Euclidean-distance clustering; it does not imply that all subjects have equal educational importance.

The sensitivity analysis repeats subject-feature selection using coverage thresholds of 70%, 80%, and 90%, while retaining the 80% student-completeness threshold. This analysis checks whether the number of students and features changes under alternative subject-coverage rules. It is not a full perturbation-based stability analysis.

C. K-Means++ Experiment

K-Means partitions n observations into k clusters by assigning each observation to its nearest centroid and iteratively updating centroids. The objective is to minimize the within-cluster sum of squared Euclidean distances. K-Means++ uses distance-aware seeding: after selecting an initial centroid, subsequent centroids are sampled with probability proportional to the squared distance from the nearest selected centroid. This initialization strategy is relevant because K-Means can converge to different local solutions depending on its initial centers, as discussed empirically by Harris and de Amorim [9].

For each cohort, the experiment evaluates $k \in \{2,3,4,5,6,7,8\}$. Each candidate uses 30 random seeds (0 - 29), init="k-means++", n_init=1, and max_iter=500. Setting n_init=1 makes each seed correspond to one explicit initialization, enabling pairwise comparison among the resulting partitions. The fixed-k reference uses conventional random initialization with k=3, n_init=10, and random_state=42. It is a baseline configuration on the same cohort matrix, not a comparison with a separate dataset.

D. Internal Validation and Initialization Repeatability

The Silhouette coefficient measures the relative cohesion and separation of a partition. For each observation, a(i) is its mean distance to observations in the same cluster and b(i) is its minimum mean distance to another cluster; the coefficient is $(b(i)-a(i))/\max(a(i),b(i))$. Higher values indicate stronger separation, values near zero indicate overlapping boundaries, and negative values indicate that an observation may be closer to another cluster.

The Davies - Bouldin index (DBI) summarizes the ratio of within-cluster dispersion to separation between cluster centroids. Lower values are preferred. Silhouette and DBI provide complementary internal views, but neither is a measure of classification accuracy or educational validity. Since the dataset has no ground-truth cluster labels, accuracy, precision, recall, and F1-score are not applicable.

The Adjusted Rand Index (ARI) compares two partitions by measuring pairwise assignment agreement adjusted for chance. For each k, ARI is calculated for all unordered pairs of the 30 seed-specific partitions. There are $30 \times 29/2 = 435$ pairwise comparisons per cohort and k. The mean pairwise ARI indicates initialization repeatability on the same fixed observations. It does not establish stability under student resampling, changes in semester, or external replication.

E. Pareto-Based Selection

The selected k is determined using three objectives: maximize mean Silhouette, maximize mean pairwise ARI, and minimize mean DBI. A candidate is Pareto-dominated when another candidate is at least as good on all three objectives and strictly better on at least one. The non-dominated candidates form the Pareto set. If multiple candidates remain, the predefined tie-break selects the highest mean Silhouette, followed by the highest mean ARI, and then the lowest mean DBI. This procedure makes the selection rule explicit and avoids relying on a single internal metric.

F. Supplementary Rival-Algorithm Benchmark

The Colab notebook additionally evaluates the algorithm families discussed by Andre et al. [10]: random-initialized K-Means, DBSCAN, BIRCH, Gaussian Mixture Model (GMM), and Mean Shift. All methods use the same standardized cohort matrices. Random K-Means tests k=2 - 8 with n_init=10 and random_state=42. DBSCAN tests $\text{eps} \in \{0.5,0.7,0.9,1.1,1.3,1.5,1.8,2.0\}$ and $\text{min_samples} \in \{5,10\}$. BIRCH tests 2 - 8 clusters with threshold=0.5. GMM tests 2 - 8 components with full covariance, $\text{reg_covar}=10^{-5}$, n_init=3, and random_state=42. Mean Shift estimates bandwidth using quantile=0.2 and at most 500 samples.

For each algorithm and cohort, valid candidate configurations are ranked by Silhouette, with lower DBI used as a secondary ordering criterion. A partition is considered invalid when fewer than two clusters remain or when every evaluated observation forms its own cluster. For DBSCAN, noise observations (label −1) are excluded from Silhouette and DBI calculations, and the noise ratio is reported separately. This handling is important: a high Silhouette calculated on a small retained subset cannot be interpreted as better overall segmentation without considering the proportion of excluded observations. Because configuration selection and metric reporting use the same matrix, the benchmark is exploratory and may be optimistic.

V. EXPERIMENTAL RESULTS

The primary 80% coverage pipeline retained 1,277 students across five cohorts. The cohort matrices contained 10 - 16 subject features. The main experiment completed 35 cohort-by-k configurations (five cohorts × seven k values), with 30

seeds and 435 pairwise ARI comparisons for each configuration.

TABLE II. COHORT FEATURE-MATRIX SUMMARY

Cohort	Students	Subjects	Imputed cells
Grade X	426	16	25
Grade XI (01 - 04)	142	13	3
Grade XI (05 - 12)	284	10	11
Grade XII MIPA	283	15	22
Grade XII IPS	142	14	25
Total	1,277	—	86

TABLE III. SELECTED K-MEANS++ RESULTS (k=2)

Cohort	N	Silhouette	DBI	Mean ARI
Grade X	426	0.3064	1.2047	1.0000
Grade XI (01 - 04)	142	0.3165	1.1787	0.9600
Grade XI (05 - 12)	284	0.2887	1.2813	0.9498
Grade XII MIPA	283	0.2624	1.4601	0.9711
Grade XII IPS	142	0.2299	1.5038	1.0000

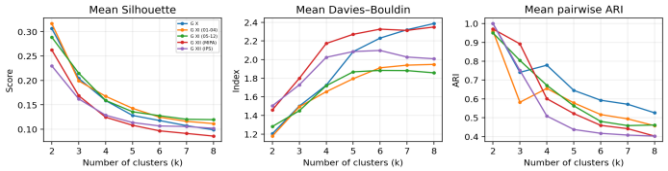


Fig. 1. Mean internal validation metrics and initialization repeatability for k=2 - 8.

Across all five cohorts, k=2 was the only non-dominated candidate under the specified Pareto procedure. The selected mean Silhouette values ranged from 0.2299 to 0.3165. The lowest DBI was 1.1787 for Grade XI classes 01 - 04, while the highest was 1.5038 for Grade XII IPS. Mean pairwise ARI ranged from 0.9498 to 1.0000. These values show that repeated K-Means++ initialization produced similar partitions for the fixed matrices, while the moderate Silhouette values indicate that the groups are not sharply separated.

Grade X and Grade XII IPS produced mean pairwise ARI of 1.0000, indicating identical partitions across the 30 initializations at k=2. Grade XI classes 01 - 04, Grade XI classes 05 - 12, and Grade XII MIPA also showed high repeatability, with mean ARI values of 0.9600, 0.9498, and 0.9711, respectively. The standard deviations for these three cohorts were 0.0416, 0.0469, and 0.0266. These results concern initialization sensitivity only and should not be generalized to different student samples.

TABLE IV. RANDOM-INITIALIZED K-MEANS REFERENCE (k=3)

Cohort	N	Silhouette	DBI
Grade X	426	0.2002	1.5269
Grade XI (01-04)	142	0.1969	1.5329
Grade XI (05-12)	284	0.2191	1.4224
Grade XII (MIPA)	283	0.1636	1.8279
Grade XII (IPS)	142	0.1623	1.7279

For every cohort, the selected K-Means++ solution at k=2 had a higher Silhouette and a lower DBI than the fixed k=3 random-initialized K-Means reference. This is a comparison between two configurations on the same feature matrix, not proof that K-Means++ is universally superior: the number of clusters differs, and the baseline uses multiple internal starts (n_init=10).

TABLE V. CLUSTER MEMBERSHIP OF SELECTED SOLUTIONS

Cohort	Lower-score	Higher-score
Grade X	201 (47.2%)	225 (52.8%)
Grade XI (01-04)	63 (44.4%)	79 (55.6%)
Grade XI (05-12)	99 (34.9%)	185 (65.1%)
Grade XII (MIPA)	107 (37.8%)	176 (62.2%)
Grade XII (IPS)	75 (52.8%)	67 (47.2%)

The selected partitions are described as lower-score and higher-score profiles according to the mean of their original-scale subject centroids within each cohort. These are relative labels, not school-defined achievement categories. The proportions vary by cohort: Grade XI classes 05 - 12 has the largest higher-score share (65.1%), whereas Grade XII IPS has a slightly larger lower-score share (52.8%).

TABLE VI. THREE LARGEST SUBJECT-CENTROID GAPS PER COHORT

Cohort	Subject	Lower	Higher	Gap
Grade X	Kimia	37.5	61.2	23.7
Grade X	Biologi	37.9	58.0	20.1
Grade X	PPKn	42.3	61.8	19.4
Grade XI (01-04)	Bahasa Indonesia	52.0	70.7	18.7
Grade XI (01-04)	Bahasa Jawa	66.3	84.8	18.5
Grade XI (01-04)	Bahasa Inggris Lanjut	31.2	48.8	17.6
Grade XI (05-12)	Penjasorkes	32.3	52.3	20.0
Grade XI (05-12)	Seni Budaya	64.5	81.0	16.5
Grade XI (05-12)	Bahasa Indonesia	53.0	69.5	16.5
Grade XII (MIPA)	Fisika	42.2	63.7	21.5
Grade XII (MIPA)	Matematika Wajib	70.0	88.5	18.6
Grade XII (MIPA)	Bahasa Inggris Wajib	30.7	48.7	18.0
Grade XII (IPS)	PPKn	36.8	55.3	18.5
Grade XII (IPS)	Bahasa Jawa	49.2	66.3	17.1
Grade XII (IPS)	Ekonomi	39.1	55.3	16.2

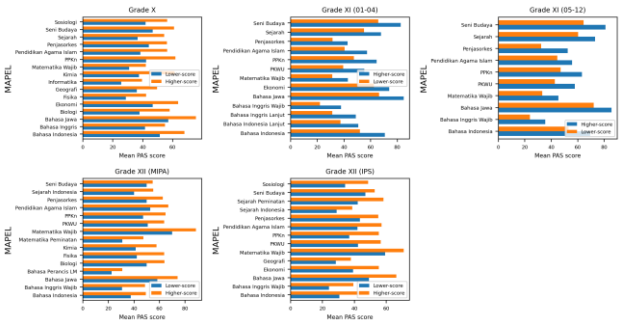


Fig. 2. Original-scale subject centroids for lower-score and higher-score profiles.

A. Coverage-Threshold Sensitivity

TABLE VII. SENSITIVITY TO SUBJECT-COVERAGE THRESHOLD

Threshold	Cohort	Students	Features
70%	Grade X	426	16
70%	Grade XI (01-04)	142	13
70%	Grade XI (05-12)	284	10
70%	Grade XII (MIPA)	283	15
70%	Grade XII (IPS)	142	15
80%	Grade X	426	16
80%	Grade XI (01-04)	142	13
80%	Grade XI (05-12)	284	10
80%	Grade XII (MIPA)	283	15
80%	Grade XII (IPS)	142	14
90%	Grade X	426	16
90%	Grade XI (01-04)	142	13
90%	Grade XI (05-12)	284	10
90%	Grade XII (MIPA)	283	15
90%	Grade XII (IPS)	142	14

Student counts remained unchanged at all three thresholds. Feature counts also remained unchanged for four cohorts. Grade XII IPS retained 15 features at 70% coverage and 14 features at both 80% and 90%. Thus, the primary 80% threshold has limited effect on the number of retained features for most cohorts, while the Grade XII IPS feature set is somewhat sensitive to the threshold.

B. Rival-Algorithm Comparison

TABLE VIII. BEST VALID RIVAL CONFIGURATION PER COHORT

Cohort	Method	Sil.	DBI	Clusters	Noise
Grade X	Random K-Means	0.306	1.205	2	0.0%
Grade X	DBSCAN	0.456	0.770	3	91.3%
Grade X	BIRCH	0.283	1.241	2	0.0%
Grade X	GMM	0.201	1.577	2	0.0%
Grade X	Mean Shift	—	—	—	—
Grade XI (01-04)	Random K-Means	0.317	1.176	2	0.0%
Grade XI (01-04)	DBSCAN	0.145	1.679	5	72.5%
Grade XI (01-04)	BIRCH	0.316	1.181	2	0.0%
Grade XI (01-04)	GMM	0.250	1.349	2	0.0%
Grade XI (01-04)	Mean Shift	—	—	—	—
Grade XI (05-12)	Random K-Means	0.292	1.273	2	0.0%
Grade XI (05-12)	DBSCAN	0.389	0.939	2	95.1%
Grade XI (05-12)	BIRCH	0.373	0.988	2	0.0%
Grade XI (05-12)	GMM	0.237	1.440	2	0.0%
Grade XI (05-12)	Mean Shift	0.338	1.235	3	0.0%
Grade XII (MIPA)	Random K-Means	0.261	1.463	2	0.0%
Grade XII (MIPA)	DBSCAN	—	—	—	—
Grade XII (MIPA)	BIRCH	0.260	1.472	2	0.0%
Grade XII (MIPA)	GMM	0.211	1.697	2	0.0%
Grade XII (MIPA)	Mean Shift	0.190	1.178	3	0.0%
Grade XII (IPS)	Random K-Means	0.230	1.504	2	0.0%
Grade XII (IPS)	DBSCAN	—	—	—	—
Grade XII (IPS)	BIRCH	0.241	1.299	2	0.0%
Grade XII (IPS)	GMM	0.131	2.054	2	0.0%
Grade XII (IPS)	Mean Shift	—	—	—	—

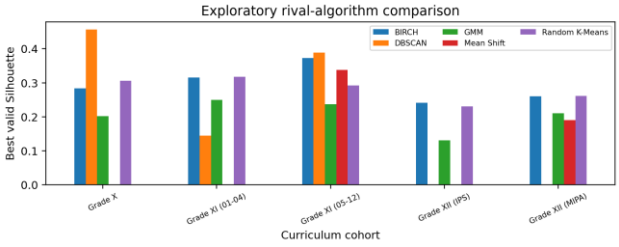


Fig. 3. Best valid Silhouette for rival methods; DBSCAN values are based only on non-noise observations.

Random K-Means and BIRCH generated valid candidates for all five cohorts. GMM also generated valid candidates for all cohorts, while Mean Shift produced a valid partition only for Grade XI classes 05 - 12 and Grade XII MIPA. DBSCAN generated valid candidates in three cohorts; no valid candidate was found for Grade XII MIPA and Grade XII IPS under the tested parameter grid.

DBSCAN's highest selected Silhouette was 0.4564 in Grade X, but it evaluated only 37 of 426 students and classified 91.3% as noise. In Grade XI classes 01 - 04, the selected DBSCAN candidate evaluated 39 of 142 students and excluded 72.5% as noise. In Grade XI classes 05 - 12, it evaluated 14 of 284 students and excluded 95.1% as noise. These large exclusion rates mean the DBSCAN Silhouette values are not directly comparable with methods that assign all students to clusters. The result is reported to expose this limitation rather than to claim DBSCAN superiority.

Across full-cohort methods, BIRCH was competitive in several cohorts. For Grade XI classes 05 - 12, BIRCH achieved a Silhouette of 0.3733 and DBI of 0.9883, exceeding the selected K-Means++ Silhouette in that cohort. For Grade XI classes 01 - 04, random K-Means reached 0.3174, close to K-Means++ at 0.3165. These exploratory comparisons suggest that algorithm performance varies by cohort. They do not establish a universal winner because the candidate selection is performed on the same data used for reporting, and the tuning budgets differ across algorithms.

VI. DISCUSSION

The principal finding is that subject-level PAS matrices can be constructed for five curriculum-compatible cohorts and yield repeatable two-cluster partitions under repeated K-Means++ initialization. This approach preserves subject-level variation that would be lost if the model used only a single aggregate statistic. It is consistent with the concern raised by Chen et al. [2] that overall grades may obscure individual characteristics. However, a richer feature

representation does not automatically guarantee pedagogical validity; the profiles must be reviewed by educators and interpreted in the context of the curriculum.

The Silhouette range of approximately 0.23 - 0.32 indicates modest separation. A stable partition can still have overlapping clusters: initialization repeatability and geometric separation are distinct properties. The high ARI values show that the optimizer repeatedly reaches similar partitions for the same data matrix, but the moderate Silhouette values caution against treating the groups as sharply distinct. This distinction is important when translating a mathematical grouping into an educational narrative.

The largest centroid gaps differ across cohorts. Grade X shows prominent gaps in Chemistry, Biology, and Civics. Grade XI classes 01 - 04 shows gaps in Indonesian Language, Javanese Language, and Advanced English. Grade XI classes 05 - 12 shows gaps in Physical Education, Arts and Culture, and Indonesian Language. Grade XII MIPA shows gaps in Physics, Compulsory Mathematics, and Compulsory English. Grade XII IPS shows gaps in Civics, Javanese Language, and Economics. These differences identify where the two profiles diverge most in the observed score matrix; they do not measure subject importance, difficulty, teacher effectiveness, or causal influence.

EDM research has used diverse data sources, including grades, online activity, programming logs, and digital traces [3], [7], [8], [11]. The present dataset contains only semester assessment outcomes. Therefore, the analysis cannot infer the learning process, student engagement, attendance, or reasons for lower scores. The results should be treated as an initial diagnostic summary that may guide further review, not as a replacement for formative assessment or teacher judgment.

The rival benchmark also shows why metric context matters. DBSCAN can produce a high Silhouette among retained core points while assigning most students to noise. Reporting only the Silhouette would hide this loss of coverage. In addition, the best rival configuration is selected using the same matrix on which its metric is reported. This creates selection optimism. A stronger comparison would define a separate tuning protocol, assess assignment coverage, repeat stochastic methods, and evaluate stability under resampling. Andre et al. [10] motivate the algorithm families but their reported results cannot be transferred to this dataset.

VII. IMPLEMENTATION AND REPRODUCIBILITY

The experiment is implemented in Google Colab using Python, pandas, NumPy, Matplotlib, and scikit-learn. The notebook follows a sequential pipeline: mount Google Drive and load the PAS workbook; audit raw records; normalize class labels; convert score values to numeric; resolve paired nonnumeric markers; assign curriculum cohorts; build subject matrices; filter subjects and students; impute missing values; remove zero-variance features; standardize; run K-Means++; calculate validation metrics; select k using Pareto dominance; profile centroids; conduct coverage sensitivity analysis; execute rival algorithms; and export aggregated results.

The primary experiment contains $5 \times 7 \times 30 = 1{,}050$ K-Means++ fits. For each cohort and k, the 30 resulting partitions generate 435 pairwise ARI comparisons. The output workbook is designed to include the audit summary, feature-matrix summary, full k-metric table, Pareto-selected solutions, k=3 reference, membership summaries, coverage sensitivity, centroid profiles, rival status, and rival candidate results. The manuscript values reported here were recomputed from the supplied Dataset.xlsx using the same preprocessing and estimator settings in the uploaded Colab notebook. The notebook itself did not contain saved execution outputs; therefore, this revision independently reran its computational logic against the supplied workbook.

VIII. THREATS TO VALIDITY, ETHICS, AND LIMITATIONS

First, the dataset represents one school and one PAS semester. The findings cannot be generalized to other schools, academic years, or assessment systems without replication. Second, the cohort mapping is inferred from class labels and observed subject offerings; the school should verify that the five cohort definitions accurately represent curriculum structure. Third, median imputation is simple and reproducible, but it may reduce within-subject variation for imputed observations. Alternative missing-data treatments may change Euclidean distances and cluster assignments.

Fourth, pairwise ARI evaluates initialization repeatability on a fixed matrix. It does not test sample-level stability, temporal persistence, or external validity. Fifth, internal metrics evaluate geometric properties rather than educational usefulness. No ground-truth labels are available, and the study therefore does not report classification accuracy, precision, recall, or F1-score. Sixth, the rival comparison uses heuristic parameter grids and selects configurations by Silhouette on the same data; it is exploratory and subject to optimistic selection. DBSCAN additionally has incomplete assignment coverage.

Student assessment records are sensitive educational information. Direct identifiers, including names and attendance numbers, must not appear in public notebooks, repositories, figures, or supplementary material. The student key is used only for record linkage and is excluded from the feature matrix. Public reporting should remain aggregated, and the original workbook should be restricted to authorized personnel. School authorization and institutional data-governance requirements must be confirmed before publication. Cluster membership should not be used as the sole basis for ranking, promotion, sanctions, or access to educational opportunities.

IX. CONCLUSION AND FUTURE WORK

This study developed a subject-level student segmentation pipeline for PAS records from SMA Negeri 1 Jakenan. The preprocessing workflow transformed 19,401 raw records into five curriculum-compatible cohorts and retained 1,277 students with 10 - 16 subject features per cohort. K-Means++ was evaluated for k=2 - 8 using 30 random initializations, and the selected k was 2 for all cohorts under Pareto selection over mean Silhouette, pairwise ARI, and mean DBI. The selected solutions achieved Silhouette values of 0.2299 - 0.3165, DBI values of 1.1787 - 1.5038, and mean pairwise ARI values of 0.9498 - 1.0000.

The findings support repeatability of the selected partitions with respect to centroid initialization, but indicate only moderate separation. Subject-level centroid gaps provide a more detailed description than a single aggregate score, while remaining descriptive and non-causal. The supplementary comparison shows that BIRCH and random K-Means can be competitive in particular cohorts, and that

DBSCAN's high Silhouette values must be interpreted alongside substantial noise exclusion. No algorithm is declared universally superior.

Future work should evaluate sample-level stability through repeated subsampling or bootstrap experiments, replicate the analysis on subsequent semesters, compare alternative imputation and distance strategies, and obtain structured feedback from subject teachers. Such extensions would test whether the profiles persist across data perturbations and whether they provide useful information for academic support. The current output is best regarded as an exploratory decision-support aid rather than an automated student assessment system.

ACKNOWLEDGMENT

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APPENDIX A. DETAILED LITERATURE SYNTHESIS

To clarify the relationship between the present study and the eleven supplied journal articles, the literature can be organized into four connected strands. The first strand concerns the scope and foundations of EDM. Dutt et al. [1] review educational data mining as a field that adapts data-mining techniques to educational settings and describe clustering as one of the methods used to identify groups of similar educational observations. Their review supports the choice of an unsupervised exploratory approach, while also underscoring that educational datasets have contextual characteristics that should be considered during preprocessing.

The second strand focuses on academic performance and outcome-oriented analytics. Rahman et al. [3] analyze practical programming skills together with academic scores, and show how unsupervised grouping can reveal profiles that may be useful for differentiated support. Alhazmi and Sheneamer [4] examine both clustering and classification in the context of early performance analysis in higher education. Qin et al. [5] develop an early-warning model integrating subjective and objective elements, and Vives et al. [6] evaluate LSTM-based prediction at multiple points in a programming course. These papers share the broad motivation of supporting academic decisions through data, but their prediction-oriented targets and higher-education/programming contexts are not equivalent to the present secondary-school PAS segmentation task.

The third strand concerns alternative data sources for understanding students. Pecuchova and Drlik [7] analyze Moodle digital traces and compare clustering approaches to identify student activity profiles related to dropout risk. Chen et al. [8] use versioned software repositories to detect learning behavior in programming assignments. Mutka et al. [11] analyze detailed process indicators such as coding activity and submission patterns to support retention-oriented prediction. These studies demonstrate that learning analytics can represent processes, engagement, and temporal behavior. In contrast, the present study has only one semester of subject assessment scores. The distinction matters because score-based clusters cannot be interpreted as behavior types, engagement levels, or dropout-risk classes.

The fourth strand concerns clustering algorithms and their empirical comparison. Harris and de Amorim [9] show that initialization algorithms can affect K-Means outcomes across a range of data conditions. Andre et al. [10] compare K-Means, DBSCAN, BIRCH, Gaussian Mixture Models, and Mean Shift in an EDM case study involving undergraduate thesis-topic data. Their work is particularly relevant to the supplementary benchmark in this paper. However, the feature space and educational problem differ: thesis-topic representations are not numeric PAS score vectors. Therefore, the present study adopts the algorithm families as a comparative design reference but does not compare its numerical metrics directly with Andre et al.'s published values.

The eleven studies collectively establish that EDM includes descriptive clustering, predictive modeling, and learning analytics based on multiple data sources. The present contribution is intentionally specific: it examines whether student-by-subject PAS vectors can be segmented within curriculum-compatible cohorts, and whether a selected K-Means++ partition is repeatable across random initializations. The novelty claim is contextual and methodological, not a claim to invent a new clustering algorithm. In particular, the study's use of a Pareto rule combines internal separation and initialization repeatability so that the selected number of clusters is not chosen solely from a single metric.

APPENDIX B. FORMAL EXPERIMENTAL DEFINITIONS

Let C denote a curriculum cohort, and let $X_C \in R^{(n_C \times p_C)}$ denote its complete, imputed student-by-subject score matrix. After standardization, Z_C is used as the input to all clustering algorithms. For K-Means, the objective for a candidate k is to minimize $J=\Sigma_{(r=1)}^{k} \Sigma_{(z_i \in S_r)} ||z_i - \mu_r||^2$, where S_r is cluster r and μ_r is its centroid. The objective is non-convex with respect to the joint assignments and centroids; consequently, the solution may depend on initialization. K-Means++ addresses the seeding stage by favoring points that are distant from existing centers, but it does not guarantee a global optimum.

For a candidate partition P, the Silhouette value for observation i is $s(i)=(b(i)-a(i))/\max\{a(i),b(i)\}$, where a(i) is the mean intra-cluster distance and b(i) is the smallest mean distance to another cluster. The cohort-level score is the mean of s(i) over all evaluated observations. The Davies - Bouldin index is computed from within-cluster scatter S_r and centroid distances M_{rs}; for each cluster, the maximum ratio $(S_r+S_s)/M_{rs}$ is selected, and the index is the mean of these maxima. Lower DBI values indicate lower relative within-cluster scatter and greater centroid separation.

Let P_u and P_v be the partitions produced by two random seeds u and v. The Adjusted Rand Index is calculated from the pair-count contingency table and adjusts the Rand agreement for its expected value under a random partition model. ARI=1 indicates identical partitions, values near zero indicate chance-level agreement, and negative values indicate less agreement than expected by chance. The experiment uses all unordered seed pairs, not a comparison against a labeled reference partition. Therefore, the ARI values in this paper are measures of repeatability across initializations and must not be described as predictive accuracy.

The Pareto selection can be stated as follows. Candidate k_a dominates k_b if $Sil(k_a) \geq Sil(k_b)$, $ARI(k_a) \geq ARI(k_b)$,

and DBI(k_a)≤DBI(k_b), with at least one strict inequality. The non-dominated set K* contains candidates not dominated by any other candidate. The final selection is argmax over K* under lexicographic ordering of (Silhouette, ARI, −DBI). This deterministic rule was applied independently to each cohort. The result k=2 in all five cohorts is therefore a result of the observed metric vectors, not a manually specified cluster count.

An important distinction is that the three criteria are not statistically independent. Silhouette and DBI both use distances in the standardized feature space, while ARI compares label assignments generated by repeated runs. The Pareto procedure is a transparent decision rule rather than a formal statistical hypothesis test. It does not provide confidence intervals for the number of clusters and should be supplemented by resampling or external validation in future work.

APPENDIX C. FULL K-MEANS++ METRIC GRID

Table IX provides the mean values for every tested k in every cohort. The table makes the selection traceable and shows the systematic change in the internal criteria as the number of clusters increases. In the current experiment, Silhouette generally declines and DBI generally increases as k increases beyond two. Mean ARI also tends to decline for larger k, although some local variation is visible. These patterns explain why k=2 forms the sole Pareto candidate for each cohort under the selected objective directions.

TABLE IX. COMPLETE MEAN METRIC GRID FOR K-MEANS++

Cohort	k	Silhouette	DBI	Pairwise ARI
G X	2	0.3064	1.2047	1.0000
G X	3	0.2041	1.5010	0.7410
G X	4	0.1587	1.7252	0.7788
G X	5	0.1283	2.0848	0.6452
G X	6	0.1178	2.2305	0.5928
G X	7	0.1074	2.3203	0.5715
G X	8	0.0988	2.3865	0.5261
G XI (01-04)	2	0.3165	1.1787	0.9600
G XI (01-04)	3	0.1992	1.4933	0.5822
G XI (01-04)	4	0.1675	1.6554	0.6555
G XI (01-04)	5	0.1428	1.7948	0.5787
G XI (01-04)	6	0.1248	1.9118	0.5168
G XI (01-04)	7	0.1162	1.9398	0.4939
G XI (01-04)	8	0.1116	1.9482	0.4575
G XI (05-12)	2	0.2887	1.2813	0.9498
G XI (05-12)	3	0.2148	1.4507	0.8041
G XI (05-12)	4	0.1590	1.7195	0.6720
G XI (05-12)	5	0.1358	1.8683	0.5629
G XI (05-12)	6	0.1278	1.8832	0.4802
G XI (05-12)	7	0.1204	1.8815	0.4581
G XI (05-12)	8	0.1197	1.8575	0.4613
G XII (MIPA)	2	0.2624	1.4601	0.9711
G XII (MIPA)	3	0.1687	1.7986	0.8922
G XII (MIPA)	4	0.1246	2.1729	0.6009
G XII (MIPA)	5	0.1081	2.2710	0.5221
G XII (MIPA)	6	0.0967	2.3264	0.4594
G XII (MIPA)	7	0.0914	2.3132	0.4417
G XII (MIPA)	8	0.0860	2.3498	0.4025
G XII (IPS)	2	0.2299	1.5038	1.0000
G XII (IPS)	3	0.1622	1.7284	0.7510
G XII (IPS)	4	0.1288	2.0242	0.5089
G XII (IPS)	5	0.1138	2.0860	0.4375
G XII (IPS)	6	0.1067	2.0977	0.4171
G XII (IPS)	7	0.1055	2.0270	0.4078
G XII (IPS)	8	0.1019	2.0085	0.4024

APPENDIX D. COHORT-LEVEL INTERPRETATION

Grade X contains 426 students represented by 16 subject features. Its selected two-cluster solution divides the cohort into 201 lower-score-profile students (47.2%) and 225 higher-score-profile students (52.8%). The three largest absolute centroid differences occur in Chemistry (23.7 score points), Biology (20.1), and Civics/PPKn (19.4). The mean Silhouette is 0.3064, the DBI is 1.2047, and the pairwise ARI is 1.0000. This is the most initialization-repeatable solution in the sense that all 30 runs produced identical partitions, but the Silhouette still indicates moderate rather than strong separation.

Grade XI classes 01 - 04 contains 142 students and 13 features. The lower-score profile includes 63 students (44.4%), while the higher-score profile includes 79 (55.6%). The largest gaps are observed in Indonesian Language (18.7), Javanese Language (18.5), and Advanced English (17.6). Its mean Silhouette of 0.3165 is the highest among the five selected solutions, and its DBI of 1.1787 is the lowest. The mean pairwise ARI of 0.9600 is high, although the nonzero standard deviation shows that a small amount of initialization variation remains.

Grade XI classes 05 - 12 contains 284 students and 10 subject features. It has 99 lower-score-profile students (34.9%) and 185 higher-score-profile students (65.1%). The largest gaps occur in Physical Education (20.0), Arts and Culture (16.5), and Indonesian Language (16.5). Its mean Silhouette is 0.2887, DBI is 1.2813, and pairwise ARI is 0.9498. This cohort has the lowest mean ARI among the selected solutions, but the value remains high. The relatively larger higher-score group should not be interpreted as evidence that the cohort is academically stronger than other cohorts because the matrices and subject combinations differ.

Grade XII MIPA contains 283 students and 15 subject features. The lower-score profile includes 107 students (37.8%), and the higher-score profile includes 176 (62.2%).

Physics has the largest centroid gap (21.5), followed by Compulsory Mathematics (18.6) and Compulsory English (18.0). The mean Silhouette is 0.2624 and DBI is 1.4601. Its mean pairwise ARI of 0.9711 indicates high initialization repeatability. The subject gaps provide a descriptive basis for teachers to inspect these curriculum areas, but they do not establish that these subjects caused differences in overall academic performance.

Grade XII IPS contains 142 students and 14 subject features. It is the only cohort in which the lower-score profile is larger: 75 students (52.8%) compared with 67 (47.2%) in the higher-score profile. The largest gaps occur in Civics/PPKn (18.5), Javanese Language (17.1), and Economics (16.2). Its Silhouette of 0.2299 is the lowest among the selected solutions and its DBI of 1.5038 is the highest, while its mean pairwise ARI is 1.0000. Thus, the partition is highly repeatable but has the weakest internal separation of the five cohorts. This is a clear example of why repeatability should not be equated with cluster quality.

APPENDIX E. DETAILED RIVAL BENCHMARK INTERPRETATION

Random-initialized K-Means provides a reference for the effect of initialization strategy. Its best configuration in every cohort was k=2 with n_init=10. The best Silhouette values were 0.3064 for Grade X, 0.3174 for Grade XI classes 01 - 04, 0.2922 for Grade XI classes 05 - 12, 0.2610 for Grade XII MIPA, and 0.2299 for Grade XII IPS. These results are close to the selected K-Means++ solutions. The small differences should not be overinterpreted because the K-Means++ result summarizes 30 single-initialization runs, whereas the random baseline uses ten initializations inside one estimator fit.

BIRCH produced valid candidates for every cohort, and its best configuration was two clusters with threshold=0.5 in all five cohorts. Its Silhouette values were 0.2833, 0.3155, 0.3733, 0.2599, and 0.2409 for Grade X, Grade XI 01 - 04, Grade XI 05 - 12, Grade XII MIPA, and Grade XII IPS, respectively. The strongest result was Grade XI 05 - 12, where BIRCH's Silhouette of 0.3733 and DBI of 0.9883 were better than the selected K-Means++ values. This suggests that a different clustering geometry may fit that cohort better under the tested configuration. It does not establish that BIRCH is preferable for all cohorts.

GMM produced valid two-to-eight component candidates in every cohort. The best candidate by Silhouette used two components for each cohort. Its best Silhouette values were 0.2014, 0.2496, 0.2369, 0.2107, and 0.1307 in the cohort order used above. In this experiment, these values were lower than those of K-Means++ and BIRCH for most cohorts. The GMM results are sensitive to covariance assumptions and initialization settings; only the full-covariance configuration with the specified regularization and three initializations was evaluated.

Mean Shift produced a valid partition only for Grade XI classes 05 - 12 and Grade XII MIPA. The selected bandwidths were estimated using a 0.2 quantile and at most 500 samples. The resulting Silhouette values were 0.3379 for Grade XI 05 - 12 and 0.1902 for Grade XII MIPA. No valid candidate was obtained for the other three cohorts under this bandwidth-estimation procedure. These failures are reported as absence of a valid partition, not as evidence that Mean Shift can never be applied to those data.

DBSCAN produced valid candidates in Grade X, Grade XI 01 - 04, and Grade XI 05 - 12. However, its best selected candidates had noise ratios of 91.3%, 72.5%, and 95.1%, respectively. For example, the Grade XI 05 - 12 candidate evaluated only 14 of 284 students. A high Silhouette computed on such a small retained subset does not describe the full cohort. Future comparisons should include coverage-adjusted criteria or explicitly define a minimum assignment-coverage requirement before a density-based solution is considered acceptable.

Across the five methods, no single algorithm dominates all cohorts and all evaluation dimensions. The benchmark is useful as a diagnostic comparison of candidate partitions under the tested parameter grids. It should not be presented as a definitive algorithm ranking. A more rigorous comparison would allocate comparable tuning budgets, use nested or held-out evaluation where possible, report runtime and coverage, repeat stochastic methods across seeds, and evaluate the stability of the resulting partitions under perturbations of the student sample.

APPENDIX F. REPRODUCIBILITY CHECKLIST AND DATA GOVERNANCE

The principal computational settings required to reproduce the experiment are: source sheet=Sheet1; coverage_subject=0.80; coverage_student=0.80; k range=2 - 8; seeds=0 - 29; initialization=k-means++; n_init=1; max_iter=500; reference K-Means initialization=random; reference k=3; reference n_init=10; reference random_state=42; and scaler=StandardScaler. The sensitivity thresholds are 0.70, 0.80, and 0.90. The rival configurations are reported in Section IV-F. These parameters should be retained with the notebook and exported result workbook.

The preprocessing audit should be rerun whenever the source workbook changes. In particular, the counts of “Belum Ikut” and “Kristen” markers, the duplicate student - subject keys, the normalized class labels, and the final number of retained students should be checked. A change in any of these elements may alter the feature matrix and should trigger a new execution of the complete pipeline. The paper should not be updated by manually editing isolated metrics while leaving figures or tables from a previous run.

The exported research artifact should contain only aggregated results and anonymized analytical outputs. Student names, attendance numbers, the KEY linkage field, and individual cluster labels should be excluded from public output. If a public repository is prepared, the workbook containing row-level data should not be committed. The repository should instead include the notebook with a secure path-loading instruction, a data dictionary, aggregated result tables, and a README describing the environment and parameter settings.

Before the profiles are used in school practice, the data owner should confirm the interpretation of nonnumeric markers and the five cohort mappings. Teachers should review whether the largest centroid gaps align with current classroom evidence. A profile should be used to initiate a conversation about learning support, not to label a student permanently. The distinction between descriptive clustering and high-stakes assessment is essential for responsible use of educational data.

APPENDIX G. EXTENDED DISCUSSION OF THE EXPERIMENTAL DESIGN

The feature representation is one of the most consequential decisions in this study. The original workbook is stored in

long format, where a student can contribute multiple rows corresponding to different subjects. Clustering directly on this structure would treat each subject record as an independent observation and would not create a student-level profile. Pivoting by student and subject changes the unit of analysis from a score record to a student. This transformation is necessary because the research question concerns similarities among students across their subject-score patterns. The student key provides row identity during this transformation, but it is not used as a numeric predictor.

The cohort separation is also central to the validity of the distance calculation. If Grade X, Grade XI, Grade XII MIPA, and Grade XII IPS were combined into one matrix, many subject columns would be structurally absent for some students because the subjects are not offered uniformly. Treating these structural absences as zeros would introduce artificial dissimilarities. Imputing a median across all grades could also mix different curriculum and assessment distributions. The five-cohort design reduces this problem by constructing subject features and imputations within groups that have more comparable subject offerings. Nevertheless, the exact cohort mapping should be verified against the official curriculum and timetable because class labels alone are an imperfect proxy for course enrollment.

The 80% subject-coverage threshold was selected as a compromise between retaining a broad set of subjects and avoiding columns with extensive missingness. The student completeness threshold similarly reduces the influence of students whose vectors would otherwise contain a large fraction of imputed values. The sensitivity analysis shows that the retained student count is unchanged across 70%, 80%, and 90% feature coverage. This occurs because the student filter is applied after feature selection and because the observed matrices are sufficiently complete for most cohorts. Grade XII IPS is the only cohort whose selected feature count changes between the 70% and 80% thresholds. The sensitivity result therefore supports the stability of the matrix dimensions, but it does not prove that every individual assignment would remain unchanged under alternative thresholds.

Median imputation is appropriate as a transparent baseline for a small number of missing cells, but it has limitations. The median is a central tendency estimate and does not account for relationships among subjects, student-level missingness mechanisms, or the reason an assessment was not completed. If missingness is associated with student performance, the imputed values may bias the resulting distances. A future study could compare cohort-median imputation with k-nearest-neighbor imputation, iterative imputation, or methods that calculate distances only over jointly observed subjects. Any alternative should be evaluated without leaking information from held-out data if the objective later changes from descriptive segmentation to prediction.

Z-score standardization gives each subject a mean of approximately zero and a standard deviation of approximately one within the cohort. This is important because the K-Means objective is based on squared Euclidean distance, and a feature with a larger numerical variance could otherwise dominate the objective. Standardization does not equalize reliability, measurement error, or curricular importance. It simply rescales observed variation. If the school uses different score ranges or assessment difficulty across subjects, a standardized score vector should be interpreted as relative position within the cohort rather than as an absolute measure of subject mastery.

The repeated-seed design isolates the sensitivity of K-Means++ to initialization while keeping the data matrix fixed. For each k, the experiment runs the estimator 30 times, with one initialization in each run. This differs from setting n_init=30 in a single estimator call: the latter retains only the best run under inertia, whereas the present design preserves the separate partitions so that pairwise ARI can be computed. The design therefore supports an explicit statement about initialization repeatability. It does not quantify uncertainty due to which students are sampled, which subjects are included, or which semester is observed.

The selection of k=2 across the five cohorts is driven by the combined metric pattern. For the selected solutions, k=2 has the highest mean Silhouette among the tested values in each cohort, the lowest or near-lowest DBI, and the strongest initialization repeatability. As k increases, the algorithm divides the feature space into progressively smaller groups, but the resulting additional partitions do not improve the three criteria in the observed data. This does not mean that two clusters are the only meaningful representation of academic performance. If a school requires more granular support categories, a different objective or a hierarchical follow-up analysis may be appropriate. Such a decision should be motivated by a defined educational use case rather than by a preference for more categories.

The fixed k=3 random-initialized K-Means baseline helps contextualize the selected solution, but it is not a controlled comparison of initialization methods alone. Both k and initialization protocol differ. To isolate initialization effects more cleanly, a future experiment should compare K-Means++ and random initialization at the same k values, use matched seed budgets, and report distributions of Silhouette, DBI, inertia, runtime, and ARI. The current random K-Means rival experiment partially addresses this by evaluating k=2 - 8 with n_init=10, but the selection procedure remains different from the repeated single-start K-Means++ design.

The rival benchmark should also be interpreted in relation to algorithm assumptions. K-Means and BIRCH produce centroid-oriented or compact groupings, while GMM models observations through a mixture of probability distributions. DBSCAN identifies dense regions and labels observations outside those regions as noise. Mean Shift estimates modes in the feature distribution and can produce a data-dependent number of clusters. Since these algorithms do not share identical assumptions, their raw Silhouette values alone cannot establish a universally fair ranking. Coverage, number of clusters, parameter sensitivity, runtime, and stability should be considered together.

DBSCAN is particularly illustrative. Its highest Silhouette values were obtained after excluding most observations as noise. If the intended application is to provide a profile for every student, a solution that assigns only a small fraction of students is operationally incomplete. One possible future protocol would require a minimum assignment coverage, for example 80% or 90%, before a DBSCAN configuration is eligible for comparison. Another option is to report a two-part result: internal quality among assigned observations and coverage across the complete cohort. This study reports the noise ratio explicitly but does not impose a minimum coverage threshold, so DBSCAN's selected scores should be read as subset metrics.

The comparison with BIRCH is also instructive. BIRCH achieved a higher Silhouette than K-Means++ for Grade XI classes 05 - 12 under threshold=0.5 and two clusters. This indicates that the geometry of that cohort may be represented differently by the two methods. However, the result is based on a single threshold setting and a candidate selection procedure that uses Silhouette on the same data. A more complete benchmark would tune BIRCH threshold and branching factor, repeat the procedure across reasonable parameter values, and assess the stability of its assignments. The current result is a useful signal for follow-up rather than a final algorithm recommendation.

The educational interpretation should remain deliberately cautious. A lower-score profile is defined only by its lower average across the subject centroids of the selected partition. It does not mean that every student in that cluster has low scores in every subject. A student may be in the lower-score profile overall and still have a relative strength in one or more subjects. Conversely, a student in the higher-score profile may have a low score in a specific subject. Therefore, subject-level centroid tables are important complements to cluster membership counts. If the school uses these outputs to plan support, it should examine the actual subject-specific pattern rather than rely only on a single cluster label.

The centroid gap is a descriptive difference between the mean scores of two clusters. It is not a standardized effect size, statistical significance test, or causal estimate. The gap may reflect a range of factors, including prior knowledge, assessment difficulty, instructional opportunity, attendance, home learning conditions, or measurement variation. None of these explanatory variables are available in the supplied PAS workbook. It would therefore be inappropriate to infer that a subject with a large gap is inherently difficult, that a teacher is ineffective, or that a cluster membership is caused by a particular student characteristic.

An operational workflow could use the profiles in three stages. First, the academic team reviews the cohort-level summary to identify subjects with relatively large centroid gaps. Second, subject teachers examine formative assessments and classroom evidence to determine whether the patterns correspond to specific learning objectives or topics. Third, the team decides whether additional diagnostic assessment, enrichment, or targeted learning support is warranted. The clustering output should inform the review, not automatically prescribe the intervention. A follow-up evaluation could then examine whether the support activity is associated with changes in later assessment outcomes, although causal claims would require a suitable study design.

An additional concern is the potential for ecological interpretation errors. The cohort centroid describes an average pattern for a group, not the experience of every individual student. Group-level differences cannot be transferred to each member without checking individual subject results. The same caution applies when comparing cohorts: each cohort uses a different feature set and is standardized independently. Thus, the Silhouette or DBI values can be compared as descriptive diagnostics, but the average original-scale centroid values should not be treated as a common standardized ranking across all grade levels.

Finally, the paper's reproducibility depends on preserving the exact source workbook version and notebook configuration. A change in class labels, duplicate resolution, nonnumeric marker handling, or score values can change the pivot matrix. The notebook should record package versions, execution date, source-file checksum, and output workbook checksum in a future revision. The current paper reports the settings that define the experiment and the numerical outputs recomputed from the attached workbook. The generated figures and tables should be regenerated whenever the source dataset or code changes.

APPENDIX H. RESEARCH SCOPE AND CLAIM BOUNDARIES

The study is descriptive and exploratory. It answers how PAS subject scores can be transformed into cohort-specific matrices, how a candidate number of clusters can be selected from internal metrics and initialization repeatability, and what broad score profiles are visible in the selected partitions. It does not answer whether a student will pass or fail a future examination, whether a student is at risk of dropping out, or whether a particular intervention will improve achievement. Those questions require labeled outcomes, longitudinal records, and an evaluation design appropriate to prediction or causal inference.

The distinction between segmentation and prediction is important because several supplied references study prediction or early warning [4] - [6], [11]. Their reported predictive metrics cannot be transferred to this paper. Similarly, the digital-trace studies [7], [8] analyze engagement or learning behavior that is not present in the PAS workbook. The present study uses the literature to establish the relevance of EDM and to motivate methodological choices, while keeping its own claims within the scope of the observed score data.

The use of the term “stability-aware” in the title refers specifically to the repeated-initialization assessment using pairwise ARI. It should not be interpreted as a claim of full stability under sampling, time, or feature perturbation. The paper explicitly separates algorithmic repeatability from broader robustness. Future revisions may strengthen the title claim by adding bootstrap or subsampling stability analysis, but those procedures are not part of the current Colab experiment.

The phrase “higher-score” and “lower-score” is used only to label the two centroids after ordering their mean original-scale subject scores. The labels are relative to the cohort and should not be translated into “high achiever” and “low achiever” categories without a school-approved benchmark. No threshold-based educational classification is included. This distinction is necessary to avoid turning an unsupervised exploratory output into an unsupported student ranking.

The main limitation of the current dataset is that it is a single assessment snapshot. The paper does not have prior semester records, formative assessment sequences, attendance, demographic context, or classroom intervention records. Consequently, the segmentation cannot reveal student trajectories, causes of performance differences, or the persistence of a profile. A longitudinal extension with harmonized subject names, curriculum versions, and grading scales would be necessary to study transitions between profiles over time.

APPENDIX I. STEPWISE PROCEDURE FOR REPRODUCIBLE EXECUTION

The execution sequence begins by loading Sheet1 from the source workbook and preserving the original fields for audit. NILAI is converted to a numeric representation using

coercion for nonnumeric entries. The class label is normalized before a composite student key is formed. The notebook then inspects repeated student - subject pairs and checks whether a numeric value is present alongside the nonnumeric “Kristen” marker. Numeric records are prioritized, and one row is retained per key and subject. This rule is deliberately explicit because duplicate handling can affect the subject matrix.

The next stage maps each normalized class to one of five cohorts. Grade X is assigned as a single cohort. Grade XI is split into classes 01 - 04 and 05 - 12. Grade XII is split into MIPA and IPS streams. The resulting cohort label is attached to each record before the pivot operation. For every cohort, the notebook creates a student-by-subject matrix, selects features meeting the coverage threshold, filters students meeting the completeness threshold, records the number of missing cells, imputes those cells with subject medians, removes zero-variance columns, and standardizes the remaining values.

For each cohort and each k from two through eight, the notebook fits 30 separate K-Means++ models. Each fit uses one initialization and a different random_state. After each fit, the model labels are retained in memory, and the Silhouette and DBI values are computed on the complete cohort matrix. The 30 label vectors are then compared pairwise using ARI. The notebook stores the mean and standard deviation of the three metrics. It also performs a fixed k=3 random-initialized reference fit, computes original-scale centroid profiles for the selected partition, and evaluates the three coverage thresholds.

The rival stage uses the same standardized matrices. It iterates through the candidate grids for random K-Means, DBSCAN, BIRCH, and GMM, and estimates a bandwidth for Mean Shift. Each candidate is checked for metric validity. For DBSCAN, noise observations are masked before the internal metrics are calculated. The selected candidate for each algorithm and cohort is the valid candidate with the highest Silhouette, with DBI used as a secondary ordering criterion. The complete candidate list and a status summary should be retained so that a missing valid result is not confused with an algorithm that was never executed.

The final stage exports aggregated tables and cohort centroid profiles. The output must not include student names, attendance numbers, composite keys, or individual cluster labels. The experiment's integrity checks verify that all five cohorts exist, that 35 primary cohort-by-k configurations are present, that each configuration contains 30 runs, and that each has 435 pairwise ARI values. The expected retained population is 1,277 students. A failed assertion should be treated as a reason to investigate the data pipeline, not bypassed merely to produce an output file.

Before a final submission, the author should execute the complete notebook in the target Google Colab environment from a clean runtime. The source workbook should be mounted from the authorized Drive location, and the notebook should be run from the first cell through the export cell without relying on stale variables. The exported workbook should then be checked against the tables in the manuscript. This end-to-end check is important because a notebook can contain correct code while an earlier output workbook or manuscript table reflects a previous version of the data.

APPENDIX J. INTERPRETATION OF INTERNAL METRIC TRADE-OFFS

The observed decrease in Silhouette as k increases is consistent with the selected two-group solution providing the strongest average cohesion and separation among the tested candidates. However, the Silhouette coefficient is an average and can conceal variation among individual observations. Some students may lie close to the decision boundary even when the cohort-level mean is positive. A future analysis could report the distribution of individual Silhouette values, including quartiles and the proportion of negative values, to provide a more nuanced description of overlap.

The Davies - Bouldin index increases for most cohorts as k increases beyond two. This pattern indicates that the additional clusters do not consistently improve the ratio between within-cluster scatter and between-centroid distance. There are minor non-monotonic changes in some cohorts at larger k, but these do not make the larger-k candidates non-dominated under the complete three-objective rule. Reporting both Silhouette and DBI is useful because one metric alone may favor a partition that has undesirable characteristics under another view of cluster geometry.

The ARI pattern is also informative. At k=2, the pairwise ARI values are very high. At larger k, they generally decline, indicating greater sensitivity to initialization when the model is asked to divide the cohort into more groups. This does not imply that the data contain exactly two natural populations. It indicates that the two-cluster partition is easier for the algorithm to recover consistently under the specified feature space and optimization settings. The interpretation is algorithmic and should remain separate from educational theory about the number of student ability levels.

Taken together, the three metrics describe a trade-off among geometric separation, relative dispersion, and initialization repeatability. Pareto selection does not collapse these dimensions into a single weighted score. Instead, it removes candidates that are unambiguously worse on all three criteria. This is preferable to assigning arbitrary weights without a substantive justification. Nevertheless, the result depends on the chosen metrics and their directions. A different educational objective—for example, identifying a small group requiring intensive support—might motivate a different selection protocol, but such an objective would need to be specified before analyzing the results.

The internal metrics are also influenced by the standardized Euclidean feature space. If the score distributions are highly skewed, contain outliers, or exhibit nonlinear relationships, a different distance measure or clustering family may be more appropriate. The current work does not test robust scaling, rank transformations, cosine distance, or feature weighting. The findings therefore apply to the exact preprocessing and distance configuration reported in this paper. This boundary should be made explicit when comparing the study with other EDM applications that use behavioral, categorical, or temporal variables.

REFERENCES

[1] A. Dutt, M. A. Ismail, and T. Herawan, “A systematic review on educational data mining,” IEEE Access, vol. 5, pp. 15991 - 16005, 2017, doi: 10.1109/ACCESS.2017.2654247.

[2] Z. Chen, G. Cen, Y. Wei, and Z. Li, “Student performance prediction approach based on educational data mining,” IEEE Access, vol. 11, pp. 131260 - 131272, 2023, doi: 10.1109/ACCESS.2023.3335985.

[3] M. M. Rahman, Y. Watanobe, R. U. Kiran, T. C. Thang, and I. Paik, “Impact of practical skills on academic performance: A data-driven analysis,” IEEE Access, vol. 9, pp. 139975 - 139993, 2021, doi: 10.1109/ACCESS.2021.3119145.

[4] E. Alhazmi and A. Sheneamer, “Early predicting of students performance in higher education,” IEEE Access, vol. 11, pp. 27579 - 27589, 2023, doi: 10.1109/ACCESS.2023.3250702.

[5] K. Qin, X. Xie, Q. He, and G. Deng, “Early warning of student performance with integration of subjective and objective elements,” IEEE Access, vol. 11, pp. 72601 - 72617, 2023, doi: 10.1109/ACCESS.2023.3295580.

[6] L. Vives, I. Cabezas, J. C. Vives, N. G. Reyes, J. Aquino, J. B. Cóndor, and S. F. Segura Altamirano, “Prediction of students’ academic performance in the programming fundamentals course using long short-term memory neural networks,” IEEE Access, vol. 12, pp. 5882 - 5898, 2024, doi: 10.1109/ACCESS.2024.3350169.

[7] J. Pecuchova and M. Drlik, “Enhancing the early student dropout prediction model through clustering analysis of students’ digital traces,” IEEE Access, vol. 12, pp. 159336 - 159367, 2024, doi: 10.1109/ACCESS.2024.3486762.

[8] J. Chen, S. Lau, J. Leinonen, V. Terragni, and N. Giacaman, “Detecting learning behavior in programming assignments by analyzing versioned repositories,” IEEE Access, vol. 12, pp. 188828 - 188844, 2024, doi: 10.1109/ACCESS.2024.3514843.

[9] S. Harris and R. C. de Amorim, “An extensive empirical comparison of k-means initialization algorithms,” IEEE Access, vol. 10, pp. 58752 - 58768, 2022, doi: 10.1109/ACCESS.2022.3179803.

[10] Andre, N. Suciati, H. Fabroyir, and E. Pardede, “Educational data mining clustering approach: Case study of undergraduate student thesis topic,” IEEE Access, vol. 11, pp. 130072 - 130088, 2023, doi: 10.1109/ACCESS.2023.3332818.

[11] A. Mutka, F. Živković Mutka, M. Žagar, and D. Tolić, “Enhancing student retention in introductory programming courses: Leveraging advanced learning validation tools and educational data mining,” IEEE Access, vol. 13, pp. 153614 - 153626, 2025, doi: 10.1109/ACCESS.2025.3604563.

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https://github.com/EZZA32/UTS_FOUNDATION_TEGAR_RAHMADANI_FIRDAUS_P36.2026.00005.git